Solutions to Mock JEE MAIN – 21 | JEE 2022

PHYSICS

1.(B) $Y = A\left[1+\cos(2kx-2\omega t)\right]$ $\left[\because\ 2\cos^2\theta = 1+\cos 2\theta\right]$

$Y - A = A\cos(2kx-2\omega t)$

$y = A\cos(2kx-2\omega t)$ [Put, Y-A = y]

$\therefore \quad A' = A$

$f = \dfrac{2\omega}{2\pi} = \dfrac{\omega}{\pi}$

2.(A) Having equal and opposite momenta, the two pieces of ice comes into rest and the loss of their kinetic energies gets converted into heat to melt it into water.
i.e., Loss in K.E. = Latent heat + specific heat

$\Rightarrow \quad 2\times\frac{1}{2}mV^2 = 2mL + 2m\,S_{ice}\Delta\theta \quad \Rightarrow \quad V = \sqrt{2\left(L+S_{ice}\Delta\theta\right)}$

$\therefore \quad V = \sqrt{2(80\times1000+0.5\times1000\times12)4.2}\,m/s = 850\,m/s$

3.(B) For isobaric process : $W = P\Delta V = nR\Delta T$

For adiabotic process : $W = -\Delta U = -nC_V\Delta T = -\frac{5}{2}nR\Delta T$

For a polytropic process $PV^x = \text{constant}$

$C = C_V - \dfrac{R}{x-1}$

Multiplying by $n\Delta T$

$nC\Delta T = nC_V\Delta T - \dfrac{nR\Delta T}{x-1}$

As $nC\Delta T = \Delta Q$ and $nC_V\Delta T = \Delta U$

$\therefore \quad W = \dfrac{-nR\Delta T}{x-1}$ (As $\Delta Q = DU + W$)

For $PV^2 =$ constant, $x = 2$

$\therefore \quad W = \dfrac{-nR\Delta T}{2-1} = -nR\Delta T$

For $\dfrac{P}{V} = \text{constant}$, $x = -1$

$\therefore \quad W = \dfrac{-nR\Delta T}{-1-1} = \dfrac{nR\Delta T}{2}$

4.(B) $W = q\vec{E}.\Delta\vec{S} = qE\hat{i}.\overrightarrow{PS}$

$= \left(qE\hat{i}\right).\left(-a\hat{i}-b\hat{j}\right) = -qEa$

5.(C) I_E (intensity due to electric field) = $\frac{1}{2}c\varepsilon_0 E^2$,

I_B (intensity due to magnetic field) = $\frac{cB^2}{2\mu_0}$

$$\frac{I_E}{I_B}=\frac{\frac{1}{2}c\varepsilon_0 E^2}{cB^2/2\mu_0}=(\varepsilon_0\mu_0)(E/B)^2=\left(\frac{1}{c^2}\right)\left(c^2\right)=1$$

$$\left(as\ c=\frac{1}{\sqrt{\varepsilon_0\mu_0}}\ and\ E/B=c\right)$$

6.(D) $E_{ph}=E_e$

$$\Rightarrow \quad \frac{hc}{\lambda_{ph}}=\frac{(h/\lambda_e)^2}{2m_e}, \text{ momentum of electron } =\frac{h}{\lambda_e}$$

$$\Rightarrow \quad \frac{c}{\lambda_{ph}}=\frac{h}{2\lambda_e^2 m_e}$$

$$\Rightarrow \quad \frac{\lambda_{ph}}{\lambda_e}=\frac{2c\lambda_e me}{h}$$

$$\lambda_e=\frac{h}{\sqrt{2m_eE_e}}$$

$$\therefore \quad \frac{\lambda_{ph}}{\lambda_e}=c\sqrt{\frac{2m_e}{E_e}}=c\sqrt{\frac{2\times 9.1\times 10^{-31}}{10^{-20}}}$$

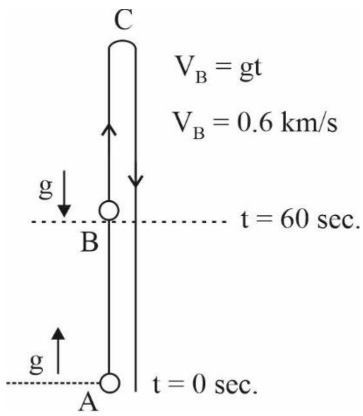
$$=1.35\times 10^{-5}\times 3\times 10^{8}=4\times 10^{3}$$

$$\lambda_{ph}>\lambda_e$$

7.(D) $\overline{AB}=\frac{1}{2}gt^2=5\times 60\times 60m=18km$

$$\overline{BC}=\frac{(600)(600)}{20}=18km,\ \overline{AC}=36km$$

$$v^2=\begin{cases} +0.02h;\ 0\le h\le 18km \\ -0.02(h-36);\ 18km\le h\le 36km \\ +0.02\,(36-h);\ 36km\ge h\ge 0km \end{cases}$$



8.(B) When the magnet approaches the coil, an induced current is set up in the coil. According to Lenz's law, a south pole is developed at the right end, so as to oppose the motion of the magnet towards the coil. The direction of the induced current as seen from magnet side of the coil is clockwise.

9.(C) y-coordinate is maximum
When $v_y=0$

$$v_y=u_y+a_yt\ \Rightarrow 0=5-5t\ \Rightarrow t=1$$

$$x = u_x t + \frac{1}{2}a_x t^2 = 0 + \frac{1}{2}(10)(1) = 5$$

$$y = u_y t + \frac{1}{2}a_y t^2 = 5(1) + \frac{1}{2}(-5)(1) = 2.5$$

10.(A) $W_{ext} = \Delta U + \Delta K$

$W_{ext} = \Delta U \qquad [\because\ \Delta K = 0]$

$= U_2 - U_1$

$= -MB\cos\theta_2 + MB\cos\theta_1 = MB\left(\cos\theta_1 - \cos\theta_2\right) = MB\left(\cos 0° - \cos 30°\right)$

$W_{ext} = MB\left(1 - \frac{\sqrt{3}}{2}\right)$

11.(D) $K_i = \frac{1}{2}mu^2$

By conservation of momentum $\qquad mu = mv_1 + nm\, v_2$

$u = v_1 + nv_2 \qquad \ldots$ (i)

$v_2 - v_1 = u_1 - u_2$

$\therefore\ v_2 - v_1 = u \qquad \ldots$ (ii)

(i) + (ii) $\qquad \Rightarrow \qquad 2u = (n+1)\, v_2$

$v_2 = \frac{2u}{n+1}$

Kinetic energy transferred $\qquad K_t = \frac{1}{2}(nm)v_2^{\,2}, \qquad K_t = \frac{2nmu^2}{(n+1)^2}, \qquad \frac{K_t}{K_i} = \frac{4n}{(n+1)^2}$

12.(D) $V = ir \qquad V = E - iR$

$i = \frac{E}{r+R}$, $\quad R =$ internal resistance

$\therefore \qquad V = \frac{Er}{r+R} \qquad \Rightarrow \qquad r + R = \frac{E}{V}r \ \Rightarrow\ R = \left(\frac{E}{V} - 1\right)r \quad \therefore \qquad R = \frac{E-V}{V}r$

13.(C) $\lambda_{\max} = \frac{hc}{10.2eV} = \frac{1240eV.nm}{10.2eV} = 121.57nm$

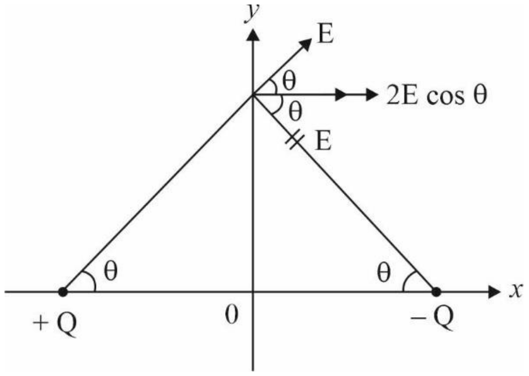
14.(C) Gravitation potential at a distance of $\frac{R}{2}$ from the centre of earth is

$$V = -\frac{GM}{2R^3}\left(3R^2 - \frac{R^2}{4}\right) = -\frac{11}{8}\frac{GM}{R}$$

Applying energy conservation

$$\frac{1}{2}mu^2 - \frac{11}{8}\frac{GMm}{R} = 0 \Rightarrow u = \sqrt{\frac{11}{4}\frac{GM}{R}}$$

15.(B) $E_x = 2E\cos\theta = 2\left(\dfrac{kQ}{a^2+y^2}\right)\left(\dfrac{a}{\sqrt{a^2+y^2}}\right) = \dfrac{Qa}{2\pi\varepsilon_0\left(a^2+y^2\right)^{3/2}}$



$y = 0,\ E_x(\max) = \dfrac{Q}{2\pi\varepsilon_0 a^2}$

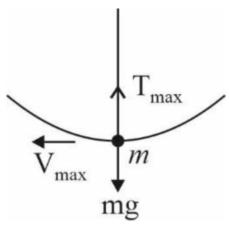
$y \to \pm\infty, E_x \to 0$

16.(C) $V_{\max} = A\omega$

$\omega = \dfrac{2\pi}{T} = \sqrt{\dfrac{g}{L}}$

$V_{\max} = A\sqrt{\dfrac{g}{L}}$ …(1)

$T_{\max} - mg = \dfrac{m(V_{\max})^2}{L}$ …(2)



By (1) and (2), $T_{\max} = mg + mg\dfrac{A^2}{L^2}$ $\therefore$ $T_{\max} = mg\left(1+\dfrac{A^2}{L^2}\right)$

17.(C) Magnetic induction due to AA´ and BB´ is given as $B_1 = \dfrac{\mu_0 i}{4\pi R}$

Magnetic induction due to CC´ and DD´ is zero at O.

Magnetic indication due to BA is given as $B_2 = \dfrac{\mu_0 i}{8R}$

Magnetic induction due to CD is given as $B_3 = \dfrac{-\mu_0 i}{8R}$

Thus net magnetic induction at O is given as $B_0 = 2B_1 + B_2 + B_3$; $B_0 = \dfrac{\mu_0 i}{2\pi R}$

18.(C) Buoyant force arises due to pressure difference between lower and upper surfaces.

$F_2 = (P_0 + \rho g R)\pi R^2$

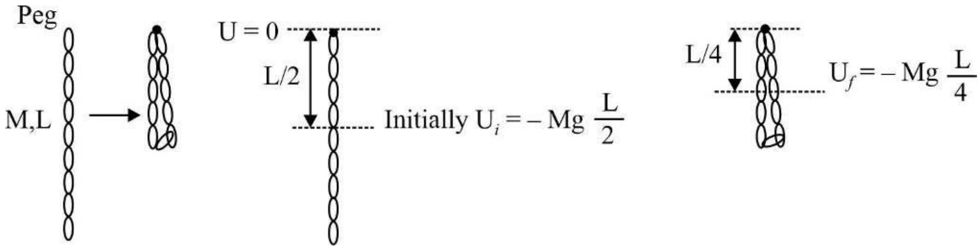
$F_2 - F_1 = \dfrac{2}{3}\pi R^3 \rho g$ [$\because$ Buoyant force = $F_2 - F_1$]

$F_1 = \left(P_0 + \rho g\dfrac{R}{3}\right)\pi R^2$ $\therefore$ $\dfrac{F_1 - P_0\pi R^2}{F_2 - P_0\pi R^2} = \dfrac{1}{3}$

19.(B) $E = e\sigma T^4 \quad \Rightarrow \quad e\sigma = ET^{-4}$

$\therefore \qquad [e][\sigma] = [E]\left[T^{-4}\right] \;\; ; \;\; [\sigma] = MT^{-3}K^{-4}$

20.(D)



Work done by external agent = Gain in Gravitational potential energy

21.(62.50)

Let the number of moles of each gas be n and let the temperature be T

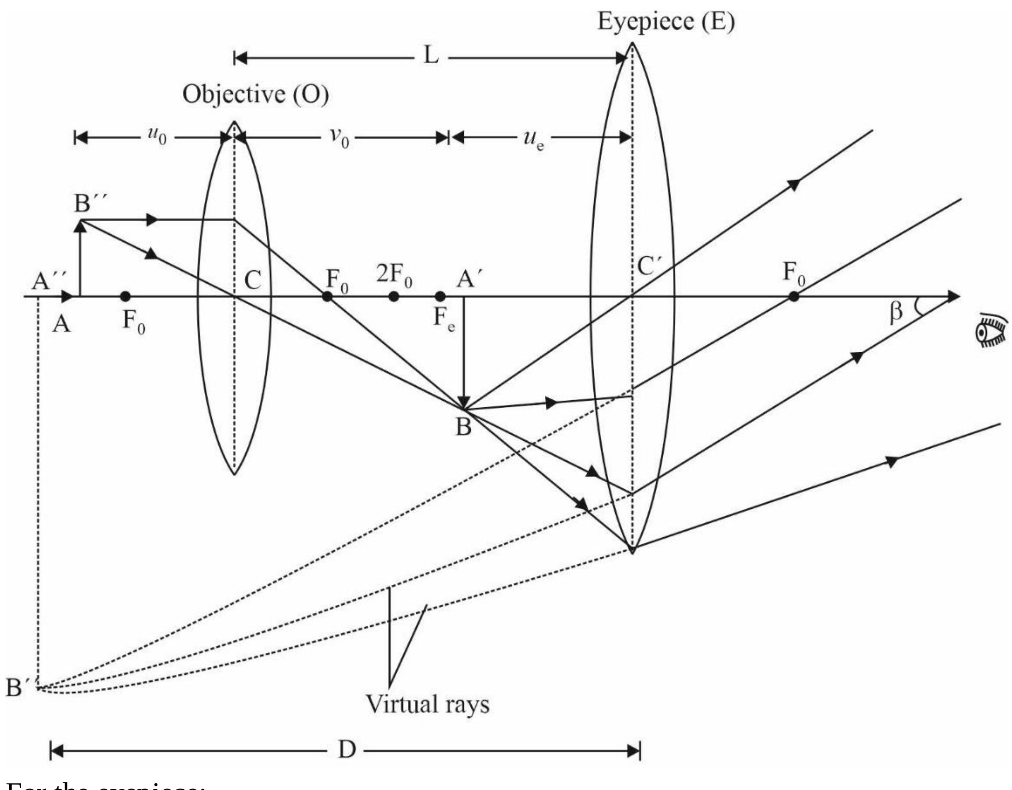
Total KE of Nitrogen molecules, $K_1 = \frac{5}{2}nRT$

Total KE of Helium molecules, $K_2 = \frac{3}{2}nRT$

Therefore, the required percentage $= \frac{K_1}{K_1 + K_2} \times 100 = 62.5$

22.(2.50) Here, $D = 25cm,\ f_0 = 2.0cm,\ f_e = 6.25cm$

$L = 15cm.$



For the eyepiece:

$$\frac{1}{f_e}=\frac{1}{v_e}-\frac{1}{u_e}, \frac{1}{u_e}=\frac{1}{v_e}-\frac{1}{f_e}=\frac{1}{-25}-\frac{1}{6.25} \qquad (\text{as } u_e=-D=-25)$$

or $\frac{1}{u_e}=-\frac{1}{5}$ *or* $u_e=-5cm$

As $v_0+|u_e|=L=15cm, v_0=15-|u_e|=15-5=10cm$

For the objective:

$$\frac{1}{f_0}=\frac{1}{v_0}-\frac{1}{u_0}$$

or $\frac{1}{2}=\frac{1}{10}-\frac{1}{u_0}$

or $\frac{1}{u_0}=\frac{1}{10}-\frac{1}{2}=-\frac{4}{10}$

or $u_0=-2.5cm$

Thus, the object should be placed at a distance 2.50 cm in front of the objective.

23.(5) Moment of inertia of the part ADE about its own centroid is $\frac{I_0}{16}$ as mass and $(\text{side})^2$ both are one-fourth of the lamina ABC

M.I. of lamina ABC about G, $I_G=3\left(\frac{I_0}{16}+Md^2\right)+\frac{I_0}{16} \Rightarrow I_0=\frac{I_0}{4}+3Md^2$

$\therefore$ $Md^2=\frac{I_0}{4}$, where M = mass of part ADE

d = distance between centroid of ADE and G.

$\therefore$ M.I. of part ADE about G = $\frac{I_0}{16}+Md^2=\frac{I_0}{16}+\frac{I_0}{4}=\frac{5I_0}{16}$ $\qquad\therefore$ N = 5.

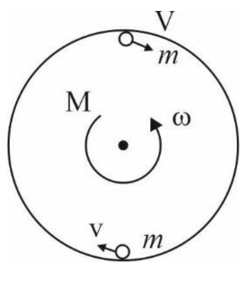
24.(1.50) $E_1=7eV-\phi_x=1eV \Rightarrow \phi_x=6eV$

$E_2=6eV-\phi_y=2eV \Rightarrow \phi_y=4eV \qquad \therefore \qquad \frac{\phi_x}{\phi_y}=\frac{6}{4}=\frac{3}{2}$

25.(2.50) v = speed relative to rim.

$v-\omega R$ = speed relative to ground.

$L_i=L_f$ as torque about axle of disc is zero.



$\Rightarrow$ $0+0+0=2m(v-\omega R)R-\frac{M}{2}R^2\omega$

$\Rightarrow$ $4mv=(4m+M)\omega R \Rightarrow 300v=750\omega R$

$\Rightarrow$ $2v=5\omega R \Rightarrow \frac{4\pi R}{t}=\frac{5\theta R}{t}$

$\therefore$ $\theta=\frac{4\pi}{5}$ radians $\qquad\therefore\qquad N=2.50$

26.(15) $I = 1000 mA = 1A$

$75V = IR + V_Z \Rightarrow 75V = IR + 60V \Rightarrow IR = 15V \qquad \therefore \qquad R = 15\Omega$

27.(15) $I_{rms} = \dfrac{V_{rms}}{X_C} = \dfrac{V_0}{\sqrt{2}} \times \omega C =$ 15 mA

28.(2) $\dfrac{\lambda_A}{\lambda_B} = \dfrac{1}{2}.$

Initial activity must be equal.

i.e., $\lambda_A N_{A_0} = \lambda_B N_{B_0}$

$\therefore \qquad \dfrac{N_{A_0}}{N_{B_0}} = \dfrac{\lambda_B}{\lambda_A} = 2$

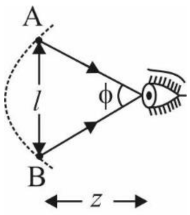
29.(3) The first polaroid eliminates half the light, so the intensity is reduced by half: $I_1 = I_0/2.$ The light reaching the second polarizer is vertically polarized and so is reduced in intensity to

$$I_2 = I_1\left(\cos 30^0\right)^2 = \frac{3}{4}I_1.$$

Thus, $I_2 = 3I_0/8.$

$\therefore \qquad n = 3$

30.(16.67) Linear distance between two dots A and B, i.e.,



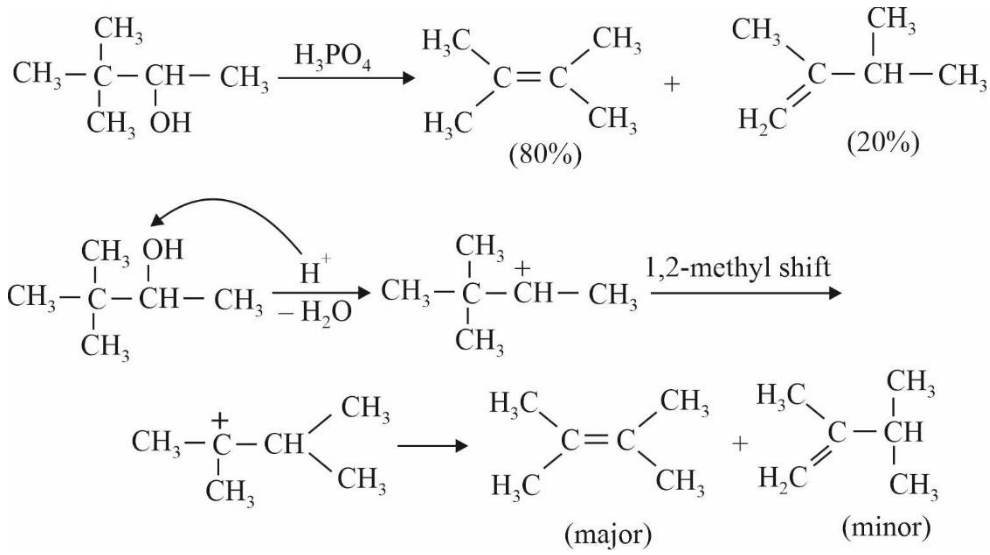
$l \approx$ length of the arc AB = $\dfrac{2.54cm}{300}$

Angular resolution of the eye, $\phi = 5.08 \times 10^{-4} rad$

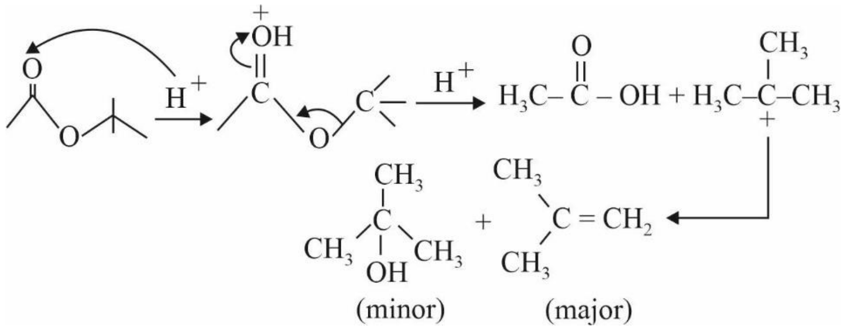
Since $\phi = \dfrac{l}{z}, z = \dfrac{l}{\phi} = \dfrac{2.54cm}{300 \times 5.08\,\text{rad}} = 16.67cm$

CHEMISTRY

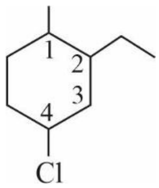
1.(B)



2.(A)



3.(C)



According to IUPAC rule, name of compound is 4-Chloro-2-ethyl-1-methylcyclohexane.

4.(B) $3Cu + 8dil.HNO_3 \rightarrow 3Cu(NO_3)_2 + \underset{(A)}{2NO} + 4H_2O$

$Fe^{2+}(aq) + NO + \underset{(conc.)}{H_2SO_4} \rightarrow \underset{(Brown)}{[Fe(H_2O)_5(NO)]^{2+}} + H_2O$

The oxidation number of Iron in $\left[Fe(H_2O)_5(NO)\right]^{2+} \Rightarrow x + (5 \times 0) + 1 = +2 \Rightarrow x = +1$

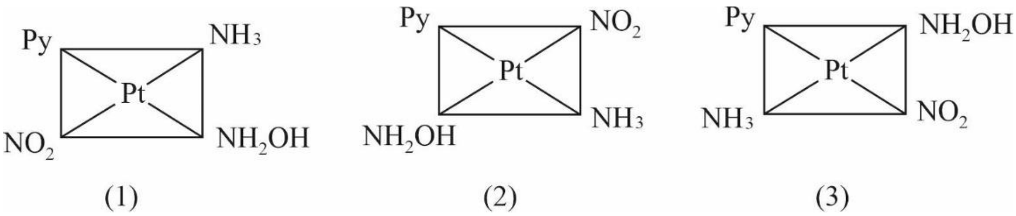
EAN of Fe = Atomic no. of metal – no. of electrons lost + no. of electrons gained from ligands
$= 26 - 1 + 12 \Rightarrow 37$

5.(D) Critical temperature of gas depends upon strength of intermolecular force of attraction. In this graph D shows more deviation from Z.
So D has highest critical temperature.

6.(C) Reactivity towards $AgNO_3$ solution depends upon the stability of carbocation intermediate. Order of stability of carbocation is (C) > (B) > (D) > (A).

7.(C) This option is correct statement if the emf. of a voltaic cell is negative, it means emf is positive for reverse reaction and reverse reaction is spontaneous, and cell is working in reverse direction.

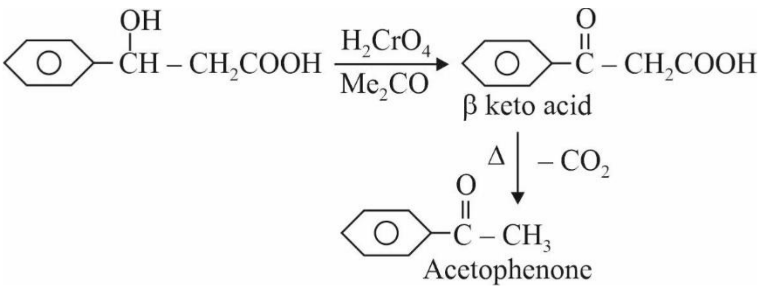
8.(C) No. of geometrical Isomers



9.(B) The increasing stability of the peroxide or superoxide, as the size of the metal ion increases, is due to the stabilisation of large anions by larger cations.

10.(C) Lysine contain $-NH_2$ group in its Zwitterion structure, so it will react with nitrous acid and liberate nitrogen gas.

11.(A)

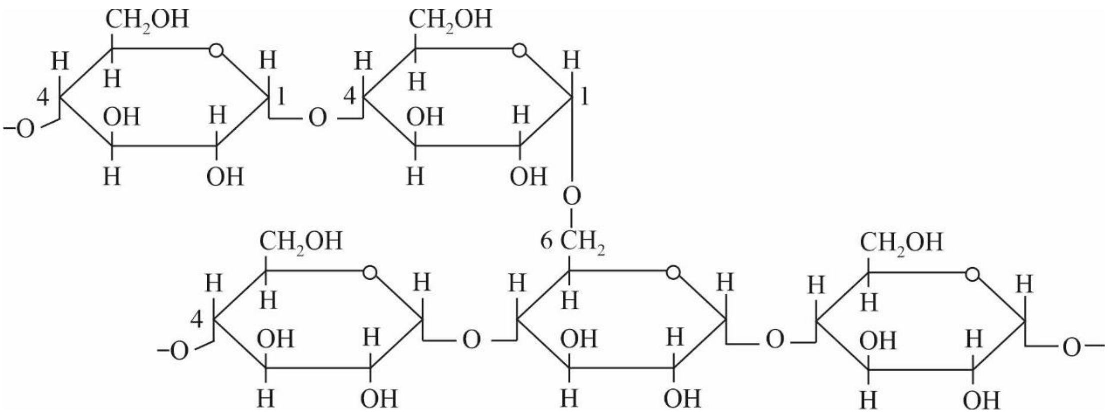


12.(A) **(Fact)**

13.(C) **(Fact)**

14.(C) **(Fact)**

15.(D)



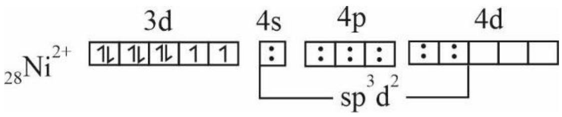
Structure of Amylopectin which make the starch insoluble.

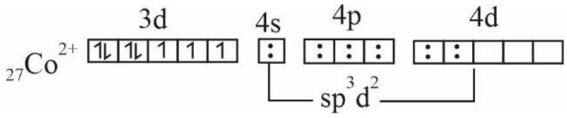
16.(D) This option is incorrect.
In isothermal process.

$$\Delta T = 0$$

And change in internal energy depend upon change in temperature. In adiabatic expansion cooling is observed,

Hence, $\Delta U = nC_V \Delta T < 0.$

17.(D) $\left[Ni\left(NH_3\right)_6\right]^{2+}$ 

$\left[Co\left(H_2O\right)_6\right]^{2+}$ 

In both complex, Ni and Co have same hybridization.

18.(B) No. of spectral lines = $\dfrac{5(5-1)}{2}=10$

19.(B) The order of Bond dissociation energy is $Cl_2 > Br_2 > F_2 > I_2$ and thermal stability depends upon Bond dissociation energy.

20.(B) At 20 min, Reaction is at the stage of equilibrium. Hence $\Delta_r G = 0$.

	R	$\rightleftharpoons$	2P	
t = 0	2		–	
t = 10	2-x		2x	(2–x = 2x)

$\Rightarrow \quad x = 2/3$

$= 4/3 \qquad = 4/3$

$$\Delta_r G^0 = -RT\ln K$$

$$= -RT\ln\left(\frac{2^2}{1}\right) = -RT\ln 4 = -\text{ ve}$$

At t = 10 min $\rightarrow \Delta_r G = \Delta_r G^0 + RT\ln Q = RT\ln\dfrac{Q}{K}$

$$= RT\ln\frac{Q}{K} = RT\ln\frac{(4/3)^2(1)}{(4/3)\times(2)^2} = RT\ln\frac{1}{3} = -\text{ve}$$

21.(5.02)

	A	$\rightarrow$	3 B
At t = 0	a		0
At t = t	$(a-x)$		3x

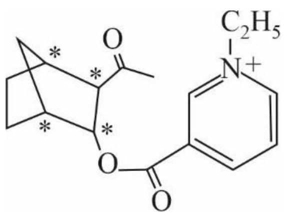
At time t, $(a-x)=3x$

$a = 4x$

$$k=\frac{2.303}{t}\log\frac{4x}{3x} \Rightarrow \frac{0.693}{t_{1/2}}=\frac{2.303}{t}\log\frac{4}{3}$$

$t_{1/2} = 5.02$ min.

22.(4)



23.(960) Moles of ethanol = $\frac{460}{46} = 10$

Moles of methanol = $\frac{x}{32}$

$\Rightarrow$ Moles fraction of ethanol = $\frac{10}{10 + \frac{x}{32}} = \frac{320}{320 + x}$

$\Rightarrow$ Vapour pressure of solution = $P_{ethanol} + P_{methanol}$

$P_T = P_A^o x_A + P_B^o x_B$

$72 = \left(\frac{320}{(320 + x)} \times 48\right) + \left(\frac{x}{(320 + x)} \times 80\right)$

$72 = \frac{(320 \times 48) + 80x}{(320 + x)}$

$x = 960$

24.(1) $\Delta T_f = iK_f \times m$

$i = \frac{\Delta T_f}{k_f \times m} = \frac{0.00732}{1.86 \times 0.0020} = 1.96 \approx 2$ Hence, complex is $\left[Co(NH_3)_4 Cl_2\right]Cl$.

25.(2.80) Meq of H_2O_2 = Meq of $KMnO_4$

$\frac{w \times 1000}{17} = \frac{0.316}{158/5} \times 1000$

$W_{H_2O_2} = 0.17\text{gm}.$

Normality of $H_2O_2 = \frac{w}{E} \times \frac{1000}{\text{Volume (in ml)}} = \frac{0.17}{17} \times \frac{1000}{20} = 0.5$

Volume strength = N × 5.6 $= 0.5 \times 5.6 = 2.8$

26.(56) $(NH_4)_2 Cr_2O_7 \xrightarrow{\text{Heat}} N_2 + 4H_2O + Cr_2O_3$

1 mole 1 mole

2 mole 2 mole

Mass of N_2 = mole of N_2 × molar mass of N_2

$= 2 \times 28 = 56$ g

27.(4) ABS Rubber, Plexiglass, Orlon, Teflon are addition polymer.

28.(5) The maximum prescribed concentration of Zn metal in drinking water is 5 ppm.

29.(4) Sodium hexametaphosphate $[Na_6P_6O_{18}]$ is known as calgon. When it is added to hard water the following reaction takes place.

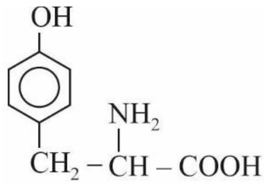
$Na_6P_6O_{18} \rightarrow 6\,Na^+ + P_6O_{18}{}^{6-}$

$3Ca^{2+} + N_6P_6O_{18} \rightarrow 6\,Na^+ + Ca_3P_6O_{18}$

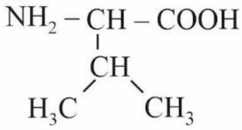
3 mole 1 mole

12 mole 4 mole

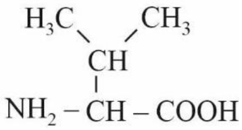
30.(3) On complete hydrolysis of the peptide give rise 3 distinct essential amino acid which are given below.



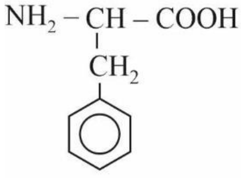
(Tyrosine)



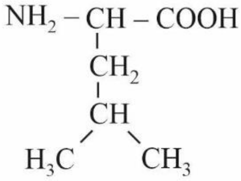
Valine



Valine



Phenylalanine



leucine

Only valine, phenylalanine and leucine are essential amino acid.

MATHEMATICS

1.(A) $\int x\frac{\left(2x+4x^3\right)}{\left(1+x^2+x^4\right)^2}dx$

$$=x\left(-\frac{1}{1+x^2+x^4}\right)-\int-\frac{1}{1+x^2+x^4}dx$$

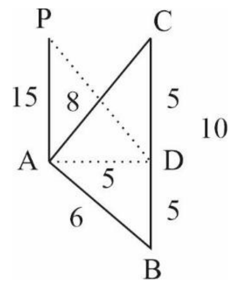
$$=-\frac{x}{1+x^2+x^4}+\frac{1}{2}\int\frac{1+x^2+1-x^2}{1+x^2+x^4}dx$$

$$=-\frac{x}{1+x^2+x^4}+\frac{1}{2}\int\frac{1+x^2}{1+x^2+x^4}dx+\frac{1}{2}\int\frac{1-x^2}{1+x^2+x^4}dx$$

$$=-\frac{x}{1+x^2+x^4}+\frac{1}{2}\int\frac{1+\frac{1}{x^2}}{\left(x-\frac{1}{x}\right)^2+3}dx+\frac{1}{2}\int\frac{(-)\left(1-\frac{1}{x^2}\right)}{\left(x+\frac{1}{x}\right)^2-1}dx$$

$$=-\frac{x}{1+x^2+x^4}+\frac{1}{2\sqrt{3}}\tan^{-1}\left(\frac{x-\frac{1}{x}}{\sqrt{3}}\right)-\frac{1}{4}\log\left(\frac{x+\frac{1}{x}-1}{x+\frac{1}{x}+1}\right)+C$$

2.(A) Midpoint of hypotenuse is circum centre which is equidistant from A, B, C.



A D = 5

$$PD=\sqrt{15^2+5^2}$$

$$PD=5\sqrt{3^2+1}=5\sqrt{10}$$

3.(C) $\sum_{r=0}^{50}\frac{{}^{100-r}C_{25}}{100-r} \Rightarrow \frac{\frac{100-r}{25}\times {}^{99-r}C_{24}}{100-r}$ $\left[{}^{n}C_r=\frac{n}{r}\times {}^{n-1}C_{r-1}\right]$

$$=\frac{1}{25}\sum_{r=0}^{50}{}^{99-r}C_{24}$$

$$\frac{1}{25}\left[{}^{99}C_{24}+{}^{98}C_{24}\ldots\ldots{}^{49}C_{24}\right]$$

$$\frac{1}{25}\left[{}^{100}C_{25}-{}^{49}C_{25}\right]$$

4.(A) $p\wedge\sim p$ is a fallacy so

$(p\wedge\sim p)\rightarrow(p\wedge q)$ is always true

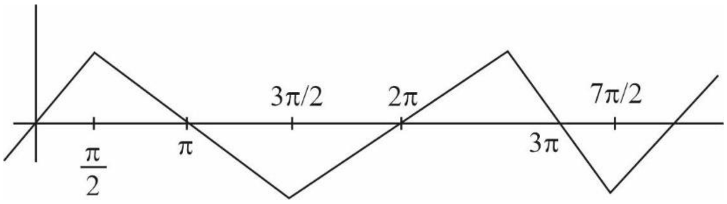
So, S_1 is tautology

5.(A) $f(x) = |\cos x| + |\sin x|$

$[f(x)] = 1$

So, $\int_{-3/2}^{3/2} 1dx = [x]_{-3/2}^{3/2} = 3$

6.(C)



7.(A) $f(x) = [\bar{a}\,\bar{b}\,\bar{c}]^2 = \begin{vmatrix} 1 & x & x^2 \\ x & x^2 & 1 \\ x^2 & 1 & x \end{vmatrix}^2$

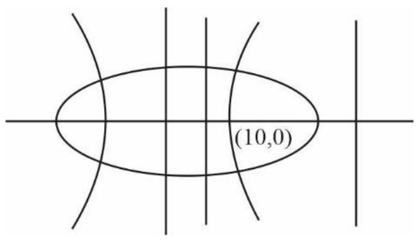
$$= (1+x+x^2)^2 \begin{vmatrix} 1 & x & x^2 \\ 1 & x^2 & 1 \\ 1 & 1 & x \end{vmatrix}^2 = (1+x+x^2)^2 \begin{vmatrix} 1 & x & x^2 \\ 0 & x^2-x & 1-x^2 \\ 0 & 1-x & x-x^2 \end{vmatrix}^2$$

$$f(x) = (1+x+x^2)^2 (1-x)^2 (1-x)^2 \begin{vmatrix} 1 & x & x^2 \\ 0 & -x & 1+x \\ 0 & 1 & x \end{vmatrix}^2$$

$$f(x) = (1-x^3)^4$$

$$f'(x) = 4(1-x^3)^3 .3x^2 = 0 \Rightarrow \begin{cases} x = 0 \\ x = 1 \end{cases}$$

8.(A) We have to find $\left|(2a_1)^2 - (2a_2)^2\right|$



$a_1e_1 = a_2e_2 = 10$

and $e_1 = \frac{10}{a_1}$; $e_2 = \frac{10}{a_2}$

also $\left|\frac{a_1}{e_1} - \frac{a_2}{e_2}\right| = 10$; $x = \frac{a_2}{e_2}$; $x = \frac{a_1}{e_1}$

$\left|\frac{a_1^2}{10} - \frac{a_2^2}{10}\right| = 10$, $\left|a_1^2 - a_2^2\right| = 100$

$\left|4a_1^2 - 4a_2^2\right| = 400$

9.(D) $f(x)=\int \frac{\left(x+1+\sqrt{x}\right)\left(x+1-\sqrt{x}\right)}{x+1+\sqrt{x}}dx$

$\int_0^x \left(x+1-\sqrt{x}\right)dx$

$f(x)=\frac{x^2}{2}+x-\frac{x^{3/2}}{3/2}$

$f(1)=\frac{1}{2}+1-\frac{2}{3}=\frac{3}{2}-\frac{2}{3}=\frac{5}{6}$

10.(B) Roots of eqn. $2x^2-3x+p=0$ are $\frac{1}{\alpha},\frac{1}{\beta}$

and of $2x^2-7x+q\ are \frac{1}{\gamma},\frac{1}{\delta}$

and given $\frac{1}{\alpha},\frac{1}{\beta},\frac{1}{\gamma},\frac{1}{\delta}$ are in AP

$\therefore \quad \frac{1}{\alpha}+\left(\frac{1}{\alpha}+d\right)=\frac{3}{2} \quad and \quad \left[\frac{1}{\alpha}+2d\right]+\left(\frac{1}{\alpha}+3d\right)=\frac{7}{2}$

$\Rightarrow \quad d=\frac{1}{2}\ and\ \frac{1}{\alpha}=\frac{1}{2}$

$\therefore \quad \frac{p}{2}=\frac{1}{\alpha}\left(\frac{1}{\alpha}+d\right)$

$\frac{p}{2}=\frac{1}{2}(1)$

$\boxed{p=1}$

$\frac{q}{2}=\frac{3}{2}\times 2$

$\boxed{q=6} \qquad \therefore \quad p+q=7$

11.(A) $\frac{a^2x}{x_1}+\frac{b^2y}{y_1}=a^2+b^2$

$\frac{x}{x_1}+\frac{y}{y_1}=2$ Passing through (6,0)

$\frac{6}{x_1}+0=2 \Rightarrow x_1=3$

$x_1^2-y_1^2=2$

$9-y_1^2=2 \Rightarrow y_1=\pm\sqrt{7}$

So Eq. of Normal is $\frac{x}{3}\pm\frac{y}{\sqrt{7}}=2$

12.(C) $d = 1$ 10, 11, 12

$d = 2$ 9, 11, 13

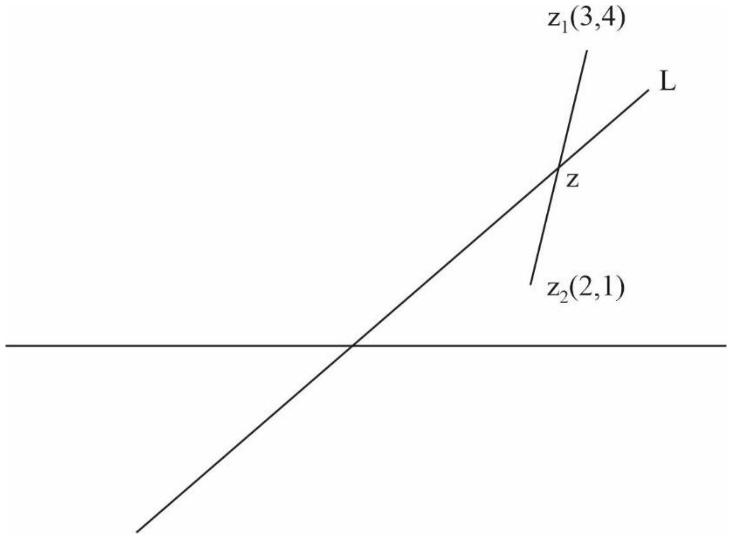
$d = 3$ 8, 11, 14

$d = 10$ 1, 11, 21

$1 + 2 + 3.....10 = 55$

13.(A) Clearly $2\left(z+\bar{z}\right)+3\left(z-\bar{z}\right)i=0$

$2x-3y=0$ (a line)



z lies on intersection of L and line joining z_1 *and* z_2 *i.e.*, $y-1=3(x-2)$

i.e., $3x-y=5$ and $2x-3y=0$

$$7x=15 \quad y=\frac{45}{7}-5=\frac{10}{7}$$

$$x=\frac{15}{7} \quad ; \quad z=x+yi=\frac{15}{7}+\frac{10}{7}i$$

14.(B) $$x+\sqrt{1+x^2}=\frac{1}{\sqrt{1+y^2}+y}=\sqrt{1+y^2}-y \qquad \ldots(1)$$

$$y+\sqrt{1+y^2}=\frac{1}{\sqrt{1+x^2}+x}=\sqrt{1+x^2}-x \qquad \ldots(2)$$

Adding (1) and (2) wegot

$x+y=0 \Rightarrow y=-x$

$$\frac{dy}{dx}=-1$$

15.(C) $A+A^2+A^3+A^4+A^5+A^6$

$$=\begin{bmatrix} \cos\theta+\cos 2\theta+\cos 3\theta+\cos 4\theta+\cos 5\theta+\cos 6\theta & \sin\theta+\sin 2\theta+\sin 3\theta+\sin 4\theta+\sin 5\theta+\sin 6\theta \\ -[\sin\theta+\sin 2\theta+\sin 3\theta+\sin 4\theta+\sin 5\theta+\sin 6\theta] & \cos\theta+\cos 2\theta+\cos 3\theta+\cos 4\theta+\cos 5\theta+\cos 6\theta \end{bmatrix}$$

$$=\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

16.(A) Let A = families own a Car
B = families own a Scooter
C = families own a Bicycle
We have to find

$\Rightarrow \quad n(A' \cap B' \cap C') = n(U) - n(A \cup B \cup C) = 100 - \left[\sum n(A) - \sum n(A \cap B) + n(A \cap B \cap C)\right]$

$= 100 - \left[(60+80+40) - 90 + x\right]$

$= 100 - [90 + x]$

$10 - x$

[Here $x = n(A \cap B \cap C) > 0$ and $0 < x \le 10$] So, possible answers is 7 %

17.(A) $\Rightarrow \left(1^2 - 2^2\right) + \left(4^2 - 5^2\right) + \left(7^2 - 8^2\right) .... + \left(3^2 + 6^2 + 9^2 ...\right)$ 30 terms

$= (1-2)(1+2) + (4-5)(4+5) + (7-8)(7+8)......$ 30 terms $+ 9\left[1^2 + 2^2 + 3^2 + ...30^2\right]$

$-1[1+2+4+5+7+8.....]$

$-[3+9+15+.....]$ 30 terms

$= -2700$ so 85095 – 2700 = 82395

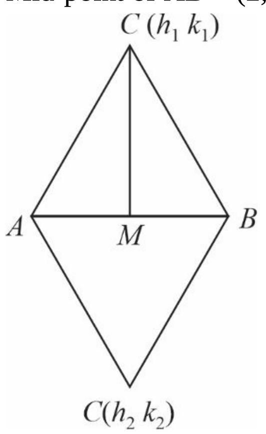
18.(A) $\dfrac{xdy - ydx}{x^2} = 2xy \quad (xdy + ydx)$

$d\left(\dfrac{y}{x}\right) = 2xy\, d(xy) \quad \Rightarrow \qquad \dfrac{y}{x} = (xy)^2 + c$

$1 = 1 + c \;\Rightarrow\; c = 0$

$\dfrac{y}{x} = (xy)^2 \Rightarrow y = \dfrac{1}{x^3} \;\; ; \;\; y(2) + y\left(\dfrac{1}{2}\right) = \dfrac{1}{2^3} + 2^3 = 8 + \dfrac{1}{8} = \dfrac{65}{8}$

19.(B) Mid-point of AB = (2, 6) (M say)



$h_1 + h_2 = 4$

$k_1 + k_2 = 12$

$\boxed{x_1 + x_2 + y_1 + y_2 = 16}$

20.(A) $\sum_{r=1}^{10} r+\sum_{r=1}^{10}\left\{\frac{r}{7}\right\}+\sum_{r=1}^{10}\left\{\frac{2r}{7}\right\}$

$=\frac{10\times 11}{2}+\underbrace{\frac{1+2+3+4+5+6+0+1+2+3}{7}}+\underbrace{\frac{2+4+6+1+3+5+0+2+4+6}{7}}$

$=55+\frac{21+6}{7}+\frac{21+12}{7}=55+\frac{27+33}{7}=55+\frac{60}{7}=\frac{385+60}{7}=\frac{445}{7}$

21.(2) Here $f(x)=\cos^{-1}x$

Now $\lim_{x\to 0}\frac{2\cos^{-1}x-\pi}{x}$ $\left(\frac{0}{0}\right)$ form LH Rule

$\Rightarrow \lim_{x\to 0}\frac{-2}{\sqrt{1-x^2}}=-2 \Rightarrow \left|\lim_{x\to 0}\frac{2f(x)-\pi}{x}\right|=+2$

22.(2) $OP=PA=PB=PC$

P is $\left(\frac{1}{2},\frac{1}{2},\frac{1}{2}\right)$

Distance of $\left(\frac{1}{2},\frac{1}{2},\frac{1}{2}\right)$ from $x+y+z-1=0$ is

$D=\frac{\frac{1}{2}+\frac{1}{2}+\frac{1}{2}-1}{\sqrt{3}}=\frac{1}{2\sqrt{3}}$

$4\sqrt{3}\,D=2$

23.(0) $\Delta=\begin{vmatrix}4&3&2\\1&-1&3\\2&5&-4\end{vmatrix}=4(4-15)-3(-4-6)+2(5+2)$

$=-44+30+14=0$

So $\Delta=0$

Now, $\Delta_1=\begin{vmatrix}1&3&2\\4&-1&3\\6&5&-4\end{vmatrix}=1(4-15)-3(-16-18)+2(20+6)$

$=-11+3\times 34+2\times 26\neq 0$

$\Delta=0, \Delta_1\neq 0$ So system is inconsistent no solution exist.

24.(7) Total no. of ways $x+y+z=15$

${}^{15+3-1}C_{3-1}={}^{17}C_2$

Favourable no of ways $x_3+y_3+z_3=6$ ${}^{6+3-1}C_{3-1}={}^{8}C_2$

$p=\frac{{}^{8}C_2}{{}^{17}C_2}=\frac{8\times 7}{17\times 16}=\frac{7}{34}$

25.(8) $a_0 = 1$

$a_{30} = 3^{15}$

$a_0 + a_1 x + a_2 x^2 + ... + a_{30} x^{30} = \left(1 + 2x + 3x^2\right)^{15}$

Diff both sides $a_1 + 2a_2 x + 3a_3 x^2 + ... = 15\left(1 + 2x + 3x^2\right)^{14}(2 + 6x)$

Put $x = 0$ we get $a_1 = 30$

$1 + 30 + 3^{15}$ digit at unit place will be 1+ 0 + 7 = 8

26.(6) $\int \sqrt{\cos^4 x + 2\sin x \cos^2 x}\, dx$

$= \int \cos x \sqrt{\cos^2 x + 2\sin x}\, dx$

$= \int \cos x \sqrt{1 + 2\sin x - \sin^2 x}\, dx = \int \sqrt{1 + 2t - t^2}\, dx = \int \sqrt{-\left[t^2 - 2t + 1 - 2\right]}\, dt$

$= \int \sqrt{2 - (1-t)^2}\, dt = \frac{1-t}{2}\sqrt{1 + 2t - t^2} + \frac{2}{2}\sin^{-1}\left(\frac{1-t}{\sqrt{2}}\right)$

$= \frac{1 - \sin x}{2}\sqrt{1 + 2\sin x - \sin^2 x} + \sin^{-1}\left(\frac{1 - \sin x}{\sqrt{2}}\right) + c$

27.(3) $\left(\frac{3}{5} + \frac{2}{5}\right)^7$ we here to get the greatest term.

Let T_{r+1} is greatest term $\left(\frac{3}{5}\right)^7 \left[1 + \frac{2}{3}\right]^7$ $x = \frac{2}{3}$

$m = \frac{|x|(n+1)}{|x|+1} = \frac{2/3(7+1)}{\frac{2}{3}+1} = \frac{16}{5}$

$r = [m] = \left[\frac{16}{5}\right] = 3$

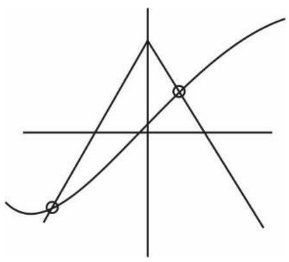
So highest winning probability that India will win 3 matches.

28.(0) Domain of $\sin^{-1} x + \cos^{-1} x + \tan^{-1} x$ is [– 1, 1]

So range is $\left[\frac{\pi}{2} - \frac{\pi}{4}, \frac{\pi}{2} + \frac{\pi}{4}\right] = \left[\frac{\pi}{4}, \frac{3\pi}{4}\right]$

$b = \frac{3\pi}{4}$

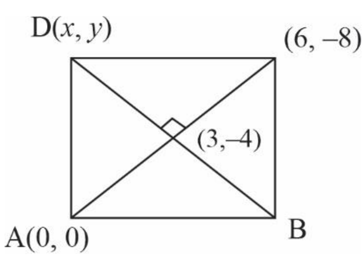
$\Rightarrow \quad [b] = 2$



Now no. of solutions of $1 - |x| = \tan^{-1} x$

2 solutions.

29.(50) Using concept of rotation



$$(x+iy)-(3-4i)=[6-8i-(3-4i)]e^{\frac{i\pi}{2}}$$

$$x+iy=(3-4i)+(3-4i)i$$

$$\Rightarrow \quad x+iy=(3-4i)(1+i)$$

$$x+iy=3-4i+3i+4$$

$$=7-i$$

$x=7, \quad y=-1$ similarly, x = – 1, y = – 7

$$7^2+1^2=50$$

30.(4)

Single digit	1-9	9
Double digit	10-99	2 × 90 = 180
Triple digit	100-999	3 × 900 = 2700
Four digit	1000-9999	4 × 9000 = 36000
Five digit	10,000	5
	9 + 180 + 2700 + 36000 + 5 = 38894	